Math 8 - Course Syllabus

Mary, Star of the Sea School | SY 2026-2027 | Dr. Peter Park

Mary, Star of the Sea School Mission

Nurture, Unite, Inspire. We are committed to nurturing each child's curiosity, uniting our classroom in respect and collaboration, and inspiring every student to grow in knowledge, faith, and wonder for the world around them.

2026-2027 Essential Question

"How can I model kindness, fairness, and humility like Jesus?"

Course Description

Math 8 at Mary, Star of the Sea is a full high-school Algebra 1 course. Algebra 1's chapters build mathematically (linear, systems, exponential, quadratic), so we follow the book's sequence closely - but we still time each chapter to match the STEM (science) calendar. Students finish the year ready for advanced high-school math placement.

Textbook: Glencoe Algebra 1

Teacher Contact

Dr. Peter Park

Email: ppark@starofthesea.org (please allow up to 24 hours for a response on school days)

Office Hours: After school from 14:45-15:15 by appointment

School office: (808) 734-0208

Materials

Provided by the school: textbook, classroom manipulatives, in-class devices.

- Composition notebook
- Pencils, erasers, highlighters
- Notebook paper
- Basic 4-function or scientific calculator
- Folder for handouts

Later in the year: graph paper, ruler/protractor, and small compass for the geometry unit.

Middle School Math Goal

From 6th through 8th grade, we focus on more than just numbers - we work to build strong math skills, boost confidence, and develop critical problem-solving abilities that students can apply in every subject and in everyday life. Our approach combines hands-on activities, real-world applications, and a supportive learning environment where students feel encouraged to take risks, ask questions, and think creatively.

By the time they graduate, our students are not only ready for Algebra and beyond, but they also possess the perseverance, analytical thinking, and resourcefulness they need to thrive in high school and in life.

Schoolwide Learning Expectations (SLEs): Self-Aware, Confident Individuals; Teachable Life-Long Learners; Active Christians; Responsible Citizens.

Course Outline by Quarter

Quarter 1 Unit 1 - Lab Math & Number Foundation

- Glencoe Ch 0 (Preparing for Algebra); Ch 7 pt. 1 (Exponents & Laws of Exponents - pulled forward)

- Lab Math Refresher: Significant Figures, Percent Error, Scientific Notation, Unit Conversions, Density review
- Real Numbers; Square Roots - supports STEM Properties of Matter Lab

  Anchor summative project: Properties of Matter Forensics Lab — *Cross-Listed with STEM*

Quarter 2 Unit 2 - Lines, Slopes & Stoichiometry

- Glencoe Ch 2 (Linear Equations); Ch 3 (Linear Functions); Ch 4 (Equations of Lines); Ch 6 (Systems)
- Slope & Rate of Change; Standard and Slope-Intercept forms; Rearranging Formulas for stoichiometry
- Percent & Weighted Averages for isotope abundance - supports Element Babybook and Stain Lab

  Anchor summative project: Element Babybook — *Cross-Listed with STEM*

Quarter 3 Unit 3 - Math for Motion

- Glencoe Ch 3-4 (Functions); Ch 10 (Radicals - Pythagorean); intro Trigonometry (beyond book)
- Physics Equations of Motion taught early January for STEM Velocity/Acceleration/Projectile Labs
- Pythagorean Theorem; SOH-CAH-TOA; Transformations; Angle Relationships

  Anchor summative project: The Apprentices Project — Marble Run Engineering — *Cross-Listed with STEM*

Quarter 4 Unit 4 - Electricity, Waves & Cosmos

- Glencoe Ch 5 (Inequalities - circuits); Ch 7 pt. 2 (Exponential, Inverse Variation); Ch 12 (Statistics & Probability)
- Ohm's Law & Linear Functions; Series and Parallel Circuits; Waves (Frequency & Wavelength - inverse variation)
- Optics & Snell's Law; Kepler's Laws of Orbital Mechanics; Volume; Scatter Plots; Two-Way Tables

Why we resequence: Algebra 1's natural order is mostly preserved - each chapter depends on the previous one. The major moves are: (1) exponent laws are pulled to Q1 for scientific-notation work, while exponential functions stay in Q4 for waves and decay; (2) physics equations of motion are front-loaded into early January for the STEM motion unit; (3) Pythagorean and basic trig are added in Q3 to support optics in Q4. Glencoe Chapters 8 (Factoring) and 9 (Quadratics) are taught in their normal place as pure-math weeks essential for high-school placement.

Math Assessment Criteria

Each criterion is scored on an MYP-style scale of 0-8. Student activities are assessed against the criteria below; end-of-quarter letter grades reflect overall progress. MYP = Middle Years International Baccalaureate Programme.

A - Knowing & Understanding. Recall and apply mathematical concepts, procedures, and formulas in both familiar and unfamiliar situations. (0-8)

B - Investigating Patterns. Apply problem-solving techniques to recognize patterns, describe them as general rules, and verify or prove those rules. (0-8)

C - Communicating. Use appropriate mathematical language (notation, symbols, terminology) in oral, written, and visual forms; move between representations; present coherent lines of reasoning. (0-8)

D - Applying Mathematics in Real-Life Contexts. Identify the relevant elements of an authentic real-world situation, select and apply an appropriate strategy to model it, and justify the accuracy and reasonableness of the result. (0-8)

Grading Policy

Tests 25% | Quizzes 25% | Classwork & Projects 30% | Participation 20%

Make-ups & Late Work: Unfinished classwork is due the next class meeting. Late work earns a maximum of 50% credit; if not turned in after two class meetings, the grade is a zero. Students below 80% on tests may make corrections/retest.

Standardized testing: Students take the STAR assessment in fall, winter, and spring to track math growth. English-language learners take the WIDA assessment as required.

Parent-teacher conferences: Formal conferences after Q1 and Q3 grades post. Additional conferences may be requested by email anytime.

How Grading Works

The four grading categories above (Participation, Tests, Classwork & Projects, Quizzes) answer what buckets your grade comes from. The MYP-style criteria (A/B/C/D, 0-8) answer how we judge the quality of your work. Both live at the same time - the MYP score for a task becomes a percentage that gets entered into one of the grading buckets.

Which criteria live in which bucket?

- Tests (25%): primarily Criterion A - Knowing & Understanding. Points earned / points possible.
- Quizzes (25%): primarily Criterion A - short-cycle knowledge and skill checks.
- Classwork & Projects (30%): where B (Investigating Patterns), C (Communicating), and D (Applying Mathematics in Real Life) live. Projects use a rubric scored 0-8 per criterion, converted to a %.
- Participation (20%): composition book, materials, engagement, safety, effort - not tied to a criterion.

Converting an MYP 0-8 score to a percentage and letter grade:

- 8 -> 95% (A)
- 7 -> 90% (A-)
- 6 -> 85% (B+)
- 5 -> 80% (B)
- 4 -> 75% (C)
- 3 -> 70% (C-)
- 2 -> 60% (D+)
- 1 -> 50% (D-)
- 0 -> 49% (F)

Worked example: a project scored A=6 and D=7. Average the 0-8 scores -> 6.5, which sits between 85% (6) and 90% (7) -> roughly 87.5% (B+). One number gets posted in Beehively under Classwork & Projects.

In plain language for families: Your quarter grade in math is calculated from four categories: Tests 25%, Quizzes 25%, Classwork & Projects 30%, and Participation 20%. Tests and quizzes are traditional point-based assessments that primarily measure Criterion A. Projects and investigations are scored using an MYP-style rubric (A/B/C/D, each 0-8) that captures deeper mathematical thinking - how well students investigate patterns (B), communicate reasoning (C), and apply math to real situations (D). Each rubric score converts to a percentage and posts in Beehively under Classwork & Projects. Participation is separate - it reflects daily readiness.

Classwork & Homework

Composition notebooks are updated every class. Any unfinished classwork is due as homework the next class. Late work: earns a maximum of 50% credit; if not turned in after two class meetings, the grade is a zero.

Classroom Procedures & Behavior Expectations

The 3 R's: Respect, Responsibility, Readiness.

Positive recognition: Verbal praise, shout-outs, positive Beehively notes home, and class-wide rewards (extra math games, free-choice problems, review-day celebrations).

If expectations aren't met: (1) private check-in and reset, (2) reflection/written response during class, (3) parent contact, (4) family meeting with the principal if needed. The goal is always to redirect and re-teach - not to punish.

Technology Use

Beehively (grade portal), Google Classroom (assignments), Freckle and IXL (targeted skill practice for differentiated practice for every unit, allowing students to strengthen their reading and language skills at their individual learning levels), Khan Academy / YouTube (supplemental video), class iPads/Chromebooks.

Digital citizenship: All device use follows the MSOS Technology Acceptable Usage Policy in the family handbook. Any misuse is handled per the school's AUP.

Student & Parent Communication

Beehively is the platform where students and parents can regularly check grade progress and receive notifications of missing or incomplete assignments.

Parents are contacted by email up to 2 times if their child is receiving a deficiency, plus standard email notifications. Students are always informed of any grade that is a C or lower and ultimately are responsible for their grades.

Student work is updated regularly.

Room Parent & Parent Volunteers

Chaperoning, guest-speaking, helping with class events (Pi Day, Math Olympics, STEM Fair). Email me - I'll connect you with our room parent. All volunteers must have current safe-environment training on file with the school office.

How to Support Your Child at Home

Consistent routines, praise for effort and strategy, weekly Beehively check with your student, real-world math conversations, and early contact if something is affecting learning.

Final Notes

Thank you for partnering with me this year. Your support at home is the single biggest factor in your child's confidence and success in math.

Class website: https://starofthesea.org/classes/

Parent Acknowledgment: After reviewing this syllabus with your student, please email ppark@starofthesea.org with the subject line "Math 8 Syllabus Reviewed - [Student Name]." Mahalo!